

# National Tsing Hua University

# NTHU DIT

## Localization

The system combines relative localization, absolute localization, and EKF fusion. Relative localization uses encoders and IMU for motion estimation, while absolute localization uses LiDAR and ArUco-based camera poses. EKF fuses both for stable positioning, balancing smoothness and drift correction. Multi-landmark redundancy improves robustness, and delayed measurements are corrected using state rollback.

## Navigation

- **Modular decision & flow control:** Uses Behavior Tree to manage navigation and task switching with action-driven real-time decisions
- **Adaptive navigation control:** Dynamically adjusts parameters and control strategies to balance speed and precision
- **Dynamic trajectory planning:** Integrates ROS2 TEB Controller for real-time optimization and obstacle avoidance
- **Vision-based alignment:** Uses ArUco markers and camera feedback for docking correction
- **Robustness & fault handling:** Combines real-time avoidance, replanning, and recovery mechanisms for higher reliability

## Vision System

1. Onboard Camera
  - **Visual Localization & Tracking:** Performs ArUco detection and PnP-based pose estimation to obtain Hazelnut relative position for real-time navigation correction and tracking
  - **Task Decision & Control:** Determines flipping actions based on marker ID and triggers control commands, while using depth sensing to evaluate manipulation success
2. External Camera
  - **Global Localization System:** Builds a Homography from four known ArUco markers to map robot image coordinates onto the world plane, and reconstructs 3D position using height information
  - **Scene Understanding & Scoring:** Uses YOLO for Hazelnut detection, integrates region masks (Collection / Pantry) for spatial counting, and computes real-time distribution and scoring for strategy support

## Mechanism

In Eurobot 2026, sheet metal is used as the chassis for rigidity and stability under high-speed motion, with improved weight distribution. Both robots share the same chassis design. The task mechanism emphasizes efficiency. PLA+ 3D printing is used for non-structural parts to reduce cost, with quick-release designs for easier maintenance. TPU is used for damping to improve stability. The system is modular, integrating wheel and PCB layout. Testing ensures reliability, with continuous optimization balancing weight, durability, and maintainability.

## Main Program

The main program acts as the decision center using an action-driven design that defines tasks without fixing execution, enabling adaptive behavior. Behavior Tree manages task flow, and DecisionCore selects targets by integrating camera, localization, opponent, and match data. A spectrum-based algorithm scores candidates by distance, value, and risk, allowing dynamic adjustments. All information is stored in a shared Blackboard, forming a consistent real-time decision loop.

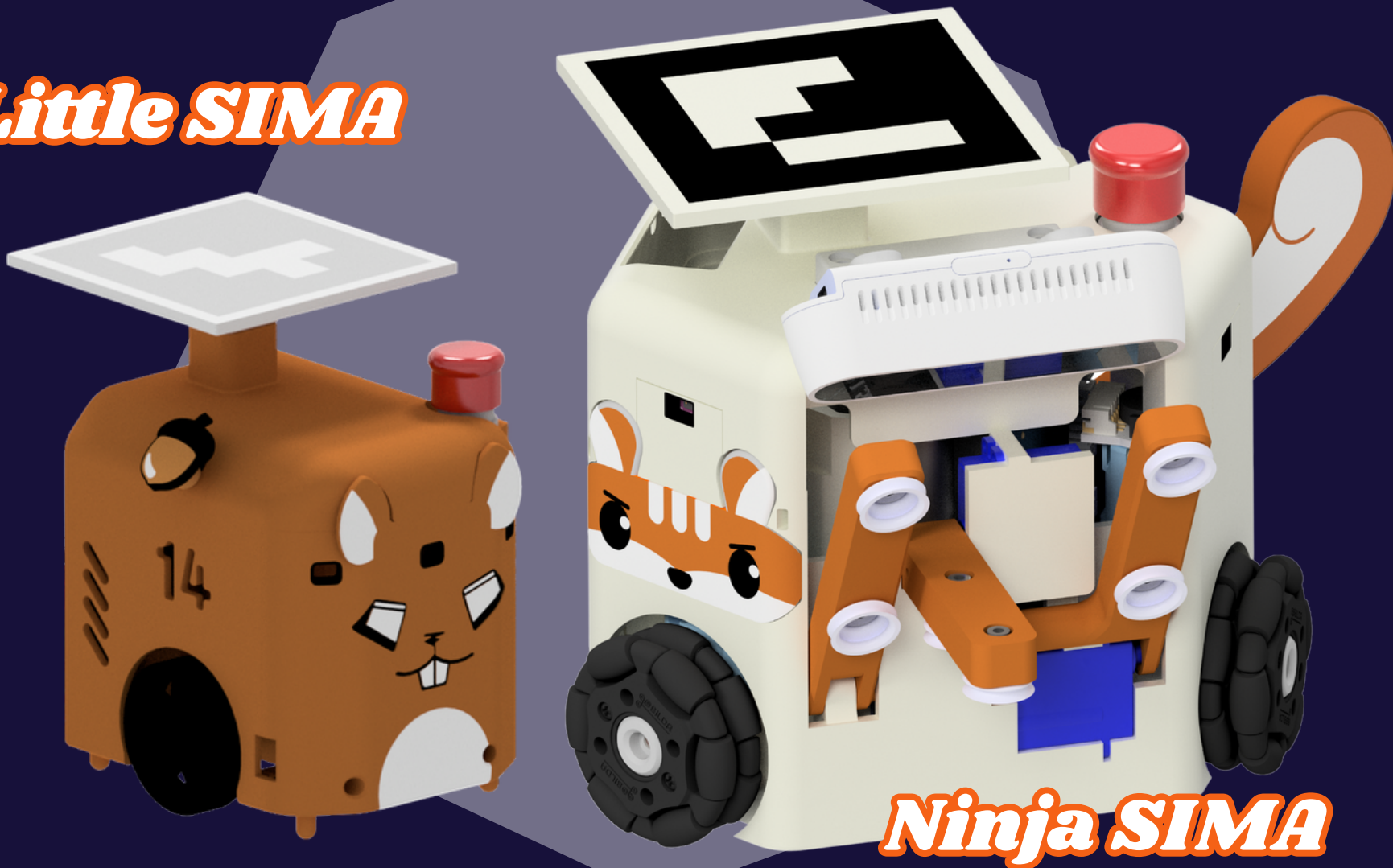
### 1. Little SIMA

- **Autonomous navigation & obstacle avoidance:** Real-time obstacle detection and avoidance using VL53L0X sensor
- **Sensor-fusion localization:** Integrates AprilTag-based global positioning with odometry for improved accuracy
- **Multi-robot collaboration:** Receives Pentry tasks and navigates to designated targets autonomously

### 2. NINJA SIMA

- **Shared navigation architecture:** Based on Little SIMA system with identical navigation and localization capability
- **Vision assistance:** Onboard camera enables Hazelnut alignment and visual recognition

## Little SIMA



## Ninja SIMA



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